Methods for comparison: how each one ranks

Source: `src/confledger/baselines/`, `ranking.py`, `evaluation_protocol.baseline_policies`, `docs/paper/comparison_methods.tex`. Call accounting is in `docs/call_accounting.pdf`.

1. The shared ranking procedure

The task is to hand over the K cases whose own answers are most likely correct (K = top 10% of a batch). Every method, including CEL-k, goes through the same four steps. Methods differ only in steps 1 and 2.

1. **Answer.** Spend the method's calls on the case and return one answer $\hat{Y}$ (usually the majority of the draws).
2. **Score.** Compute a *score key* for the case: a number, or a short tuple compared left to right (the second entry only breaks ties in the first). Larger means more confident.
3. **Sort and pick.** Sort the batch by key, descending, and take the first K. Cases with equal keys are tied; ties at the cut-off are broken uniformly at random with a fixed seed.
4. **Score the pick.** Precision@10% is the share of the K picked cases whose *own* answer $\hat{Y}$ is correct ($Z = 1\{\hat{Y} = Y\}$). AUROC and AUARC use the same keys over the whole batch. Gold labels appear only here.

Notation: a case has n usable draws; m of them agree with the majority answer, so m/n is the agreement share; $c_i \in [0, 100]$ is the verbalized confidence on draw i (free in the main configuration). A failed extraction counts as a draw that disagrees.

2. What each method returns and ranks by

Method	Draws and calls	Answer returned	Score key (larger = ranked higher)	Why it should rank well
Random	1 draw	that draw's answer	none: all cases tie, so the order is random	chance level; exact hypergeometric null
Verbalized	1 draw	that draw's answer	(c), the stated confidence; equal c ties	the model's own claim of confidence
Self-consistency	k same-prompt draws	majority	(m/n)	answers that survive resampling are more often right
Self-consistency + verbal	k same-prompt draws	majority	$(m/n,\ \bar{c})$, $\bar{c}$ = mean c_i	control: same draws, but also reads the free confidences

Method	Draws and calls	Answer returned	Score key (larger = ranked higher)	Why it should rank well
Semantic entropy	$k-1$ draws, 1 judge call	majority	$(-H,\ m/n)$, H = entropy of the reason groups	consistent *reasoning*, not only consistent answers
BSDetector	$k-1$ draws	majority	$(0.7\,m/n + 0.3\,c_1/100)$	blends observed consistency with stated confidence
Avg-Conf (self-random)	k same-prompt draws	majority $\tilde{Y}$	$(A,\ m/n)$, $A = \sum_{i:\,\hat{Y}_i=\tilde{Y}} c_i \,/\, \sum_i c_i$	confidence mass that backs the majority
Avg-Conf (prompting)	1 generator call, $k-1$ draws, one paraphrase each	majority $\tilde{Y}$	$(A,\ m/n)$	same, with varied wording
CISC	k same-prompt draws	confidence-weighted vote	(W), W = weight share of the winning answer	high-confidence draws count more in the vote and the score
Black-box calibrator	k draws; from $k=5$ add 1 generator call and paraphrase draws	majority over all draws	$(p,\ m/n)$, p = fitted $P(Z=1)$	learned map from agreement and wording similarity to correctness
CEL-k	plan chosen on dev, mean $\le k$ calls: screen, then more evidence for survivors	majority of all draws used	(s), $s = \hat{\mu} - \hat{\lambda}\hat{\sigma}$ from a fitted score model (Section 4)	combines stability, probes, challenge, entropy and free text signals, and discounts thin evidence

3. Mechanism of each score

Random. The case's single answer is returned, and the key is a constant, so the picked set is a uniform sample of the batch. Its precision is the base accuracy.

Verbalized. One response gives an answer and a stated confidence c. Cases are sorted by c. Models state coarse values (many 90s and 95s), so large tie groups form and are broken at random.

Self-consistency. Sample k draws of the original prompt. The answer is the majority, the score is m/n: a case where all k draws agree ranks above one where $k-1$ do. With k draws there are only k distinct scores, so ties are common at small k.

Self-consistency + verbal. Same draws and the same answer. It ranks by m/n first and uses the mean confidence $\bar{c}$ only to order cases with equal m/n. This is the fair control, because CEL also reads the free confidences.

Semantic entropy. Take $k-1$ same-prompt draws, each with a rationale. One judge call groups the rationales by underlying reason and returns a group label per draw. H is the Miller–Madow entropy of the label counts: 0 if all rationales give the same reason, larger if they scatter. The key is $-H$, so consistent reasoning ranks first. Answer-level ties are broken by m/n. It needs at least two draws.

BSDetector. A fixed blend: $0.7 \cdot m/n + 0.3 \cdot c_1/100$, where c_1 is the stated confidence of the case's verbalized response. We reconstruct it from cached draws, with c_1 standing in for the original self-reflection prompt. The weight 0.7 is our choice and is not fitted.

Avg-Conf. Take the majority answer $\tilde{Y}$. A is the share of the total stated confidence carried by the draws that agree with $\tilde{Y}$. Unlike m/n it weights each agreeing draw by how sure it was, so three confident votes for $\tilde{Y}$ can beat four hesitant ones. Ties are broken by m/n. The two variants differ only in how the k samples are produced: same prompt (self-random) or one paraphrase per sample (prompting).

CISC. Each draw gets weight $w_i = \mathrm{softmax}(c_i/T)$ over the case's draws. The answer is the one with the largest total weight, and the score W is that total, scaled down for failed extractions. T is fitted on the development slice to maximise the vote's accuracy over 80 values in $[10^{-4}, 10^{4}]$. A large T gives equal weights, which is self-consistency. A small T lets the single most confident draw decide. CISC defines an answer rather than a selection score, so ranking by W is our adaptation, and its answer can differ from the majority.

Black-box calibrator. For each block of draws it computes two features: the number of distinct answers and the mean pairwise ROUGE-L of the full responses (answer plus explanation). Below $k = 5$ there is one block (same-prompt draws). From $k = 5$ there are two (same-prompt and paraphrase draws), giving four features. The features are standardised, and an L2 logistic regression of Z on them is fitted on the development slice. The key is the predicted probability p, ties by m/n, and the answer is the majority over all draws.

4. CEL-k: how its score is produced

CEL-k differs from the baselines in two ways. It spends its calls unevenly (a cheap screen, then more evidence only for cases that pass), and its score is a *fitted* estimate of $P(Z = 1)$ from many signals, discounted by how thin the evidence is. Source: `methods.cel_alloc`, `score_model.py`, `docs/method.md` §2.

Step 0: choose the plan (dev slice, once per budget k). A plan is $a = (k_1, \mathrm{para}, e, F, \mathrm{chal}, \mathrm{se})$: screen size, whether the screen uses fresh paraphrases, extra draws e, probes F, challenge, rationale entropy. Among plans whose expected cost is within one call of k, take the one with the best dev prec@K. Nothing here touches evaluation items.

Step 1: run the plan on each case.

- *Screen:* para generator call plus k_1 draws. If the draws do not all agree the case is *eliminated*: it stops, its answer is the screen majority, and it keeps only the screen's evidence.
- *Survivors* (unanimous screen) buy e more draws, the probes (negation, relabel, distractor: pass/fail), and optionally the challenge and the entropy call. The answer is the majority of all draws used.

Step 2: turn the evidence into a feature row x. A signal that was not bought is marked missing.

Feature	Meaning	Costs a call?
intercept	constant	no
η (stability)	expected log-odds of the case's agreement rate: $\psi(\frac{1}{2}+m) - \psi(\frac{1}{2}+n-m)$ for m of n draws agreeing (Jeffreys prior)	no
survived	1 if the case passed the screen	no
neg, relabel, distractor	probe passed $= +1$, failed $= -1$	yes
chal	challenge held $= +1$, flipped $= -1$	yes
verb	$(\bar{c} - 50)/50$, mean stated confidence of the draws used	no
$-$se	negative rationale entropy	yes (judge)
length, hedge, neg_drop	log words in rationales, share of hedged rationales, confidence lost under negation	no
p_first	mean probability of the answer token; gray-box, only if `scoring.gray_box`	no

Step 3: the score model gives $\hat{\mu}$. On the labelled fitting split $\mathcal{D}_{\mathrm{fit}}$, replay the same plan and fit a ridge logistic regression of $Z = 1\{\hat{Y} = Y\}$ on x. This is the only place labels enter. The coefficient of η is unpenalised and constrained to be ≥ 0 (more agreement may not lower confidence). Then

$$\hat{\mu}(E) = \mathrm{sigmoid}(\hat{\beta}^{\top} x),$$

with any missing feature filled by its $\mathcal{D}_{\text{fit}}$ mean. $\hat{\mu}$ estimates $P(Z = 1 \mid E)$.

Step 4: the uncertainty $\hat{\sigma}$. The log-odds variance of $\hat{\mu}$ adds three parts:

1. fit uncertainty: $x^\top \hat{\Sigma} x$, from the inverse penalised Hessian (delta method);
2. stability uncertainty: $\hat{\beta}_\eta^2 \operatorname{Var}(\eta)$, which shrinks like $1/n$ as a case gets more draws;
3. unbought signals: $\sum_{g \text{ missing}} \hat{\beta}_g^2 \operatorname{Var}(x_g)$, using $\mathcal{D}_{\text{fit}}$ variances.

Then $\hat{\sigma} = \hat{\mu}(1 - \hat{\mu})\sqrt{\text{variance}}$. A case with fewer calls has more missing signals and fewer draws, so a larger $\hat{\sigma}$.

Step 5: the ranking score.

$$s = \hat{\mu} - \hat{\lambda}\,\hat{\sigma},$$

with $\hat{\lambda} \in \{0, 0.5, 1, 1.5, 2, 3\}$ chosen by 5-fold cross-validation on $\mathcal{D}_{\text{fit}}$ to maximise held-out prec@K. Ties go to the smaller $\hat{\lambda}$, so $\hat{\lambda} = 0$ (rank by $\hat{\mu}$) is allowed. The batch is sorted by s and the top K delivered, with random tie-breaks. There is no hard rule that eliminated cases rank below survivors; it follows from their lower η, the survived flag, and larger $\hat{\sigma}$.

What is exact and what is not. The score is model-based: nothing in it is a guarantee. When every case has the same evidence, $\hat{\sigma}$ is nearly a function of $\hat{\mu}$ and $\hat{\lambda}$ cannot reorder the top K; the pessimism only matters when evidence is uneven, as it is here. Guarantees come from the optional conformal layer applied on top of the frozen score.

Example of two feature rows (stability terms computed; coefficients are fitted, so they are left symbolic).

	Case A: survivor	Case B: eliminated
Draws	screen 3 of 3 agree, then 3 more all agree: $m = n = 6$	screen 2 of 3 agree: $m = 2$, $n = 3$
η, $\operatorname{Var}(\eta)$	3.76, 5.10	0.67, 1.43
survived	1	0
probes, challenge, SE	bought, entered as ± 1 and $-$se	not bought: filled with $\mathcal{D}_{\text{fit}}$ means; each adds $\hat{\beta}_g^2 \operatorname{Var}(x_g)$ to $\hat{\sigma}^2$
Result	large $\hat{\mu}$, small extra variance	small $\hat{\mu}$, large $\hat{\sigma}$

5. Worked example (baselines)

One case, five same-prompt draws. Answers: Yes, Yes, Yes, No, Yes. Confidences: 90, 80, 95, 60, 85. The majority is Yes, so $m/n = 4/5 = 0.8$ and $\bar{c} = 82$.

Method	Computation	Key
Self-consistency	$4/5$	0.800
Self-consistency + verbal	$(4/5,\ 82)$	(0.800, 82)
Avg-Conf	$(90 + 80 + 95 + 85) \,/\, (90 + 80 + 95 + 60 + 85) = 350/410$	(0.854, 0.800)
BSDetector, $c_1 = 90$	$0.7 \cdot 0.8 + 0.3 \cdot 0.90$	0.830

A second case with draws Yes, Yes, Yes, Yes, No and confidences 70, 65, 70, 60, 90 has the same m/n, so self-consistency ties the two. Self-consistency + verbal puts it below (mean 71 vs 82), and Avg-Conf gives $265/355 = 0.746$, also below. That is the extra information the confidence-aware methods use.

6. Departures from the source papers

- On a yes/no answer the number of distinct answers is 1 or 2, and ROUGE over answers alone is a function of m, so the calibrator computes ROUGE over full responses to keep the explanation's wording in play. We call it a black-box calibrator, not Pedapati et al.'s method, because it omits the perturbations that need new prompts or a local NLI model (sentence permutation, entity-frequency amplification, stop-word removal, split-response consistency).
- We omit Xiong et al.'s misleading-hint sampling and Top-K and Pair-Rank aggregation (Pair-Rank reduces to the top-2 probabilities on a binary question), and CISC's P(True) confidence, since all need new prompts.
- CISC with the answer token's probability in place of the length-normalised response probability is a gray-box variant, reported separately from the black-box comparisons.

7. Open check: BSDetector's call count

`bsdetector_like` (`baselines/others.py:30`) charges its draws plus one reflection call, but the reflection call is 0 when `verbal_per_draw` is on, and `baseline_policies` passes $B - 1$ draws expecting a total of B. In the main configuration BSDetector therefore spends $k - 1$ calls, one fewer than the other baselines. With `verbal_per_draw` off it spends k. Either state this in the paper or pass B draws when it is on.